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Assignment-1

SUBJECT:

Artifical Intellengce

SUBMITTED TO:

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Task#1

- Train a Machine Learning Model

Student Placement Prediction Using Machine Learning

Abstract

This project focuses on predicting student placement status using Machine Learning techniques. The main purpose of this study is to analyze student academic and behavioral factors to determine whether a student is likely to be placed or not placed. A dataset containing 10,000 student records was used for model training and evaluation. Various features such as study hours, attendance, sleep hours, internet usage, assignments completed, and previous scores were used to predict placement status. Three machine learning models were implemented: Decision Tree, Random Forest, and Improved Random Forest. The results showed that the Random Forest and Improved Random Forest models achieved the highest accuracy of 89%, while the Decision Tree model achieved 84% accuracy. This project demonstrates that machine learning can effectively predict student placement outcomes and help educational institutions identify students who may require additional support.

Introduction

Machine Learning is a branch of Artificial Intelligence (AI) that enables systems to learn patterns from data and make predictions without being explicitly programmed. In the education sector, machine learning can be used to analyze student performance and predict future academic or career outcomes.

Student placement prediction is an important task because it helps institutions understand which students are more likely to secure employment opportunities. Factors such as academic performance, attendance, study habits, and assignment completion significantly influence placement chances. By analyzing these factors, machine learning models can provide predictions about placement status.

This project applies machine learning algorithms to predict whether a student will be placed or not placed based on academic and behavioral data.

Problem Statement

Educational institutions often face difficulties in identifying students who are likely to be placed in jobs after graduation. Traditional methods of evaluation may not accurately predict placement

outcomes. Therefore, there is a need for an intelligent system that can analyze student-related factors and predict placement status effectively.

The main problem addressed in this project is:

"How can machine learning techniques be used to predict student placement status accurately using academic and behavioral factors?"

Objectives

The main objectives of this project are:

1. To analyze student-related academic and behavioral data.
2. To preprocess and prepare the dataset for machine learning.
3. To train machine learning models for placement prediction.
4. To compare the performance of different machine learning models.
5. To identify the most accurate model for predicting placement status.
6. To improve prediction accuracy through model optimization

Dataset Description

The dataset used in this project contains **10,000 student records** and **8 columns**.

Dataset Features:

study hours	No. of hours a student studies
Attendance	Student attendance percentage
Sleep hour	No. of sleeping hours
Internet usage	Daily internet usage
Assignment completed	No. of assignment completed
Previous score	Students previous academics performance
Exam score	Current exam score
Placement status	Target variables(placed/not placed)

Initially, the feature **exam_score** was removed from model training because it was directly influencing the target variable and causing unrealistic accuracy (100%). Removing this feature helped achieve more realistic and reliable results.

The target variable in this dataset is **placement_status**, which contains two categories:

- **Placed**
- **Not Placed**

Methodology

The following methodology was used in this project:

1. Data Collection

The student placement dataset containing 10,000 records was collected and imported into Google Colab from kaggle.

2. Data Preprocessing

The dataset was checked for missing values, duplicate entries, and categorical data encoding.

3. Data Splitting

The dataset was divided into training and testing sets:

- **80% Training Data**
- **20% Testing Data**

4. Model Training

Three machine learning models were trained:

- Decision Tree
- Random Forest
- Improved Random Forest

5. Model Evaluation

The models were evaluated using:

- Accuracy Score
- Confusion Matrix
- Classification Report

Preprocessing

Data preprocessing is an essential step in machine learning because raw data often contains inconsistencies.

The following preprocessing steps were performed:

Missing Values Check

The dataset was checked for missing values, and no missing values were found.

Feature Selection

The feature **exam_score** was removed because it caused data leakage and unrealistic predictions.

Label Encoding

The target variable (**placement_status**) was converted into numerical values using Label Encoding:

- Placed = 1
- Not Placed = 0

Train-Test Split

The dataset was divided into training and testing data using an **80:20 ratio**.

Model Training

Three machine learning models were implemented:

1. Decision Tree

Decision Tree is a supervised machine learning algorithm that creates a tree-like structure for classification.

Accuracy Achieved: 84%

Although Decision Tree performed well, its accuracy was lower compared to Random Forest.

2. Random Forest

Random Forest is an ensemble learning technique that combines multiple decision trees to improve performance and reduce overfitting.

Accuracy Achieved: 89%

Random Forest performed better than the Decision Tree model and produced more reliable predictions.

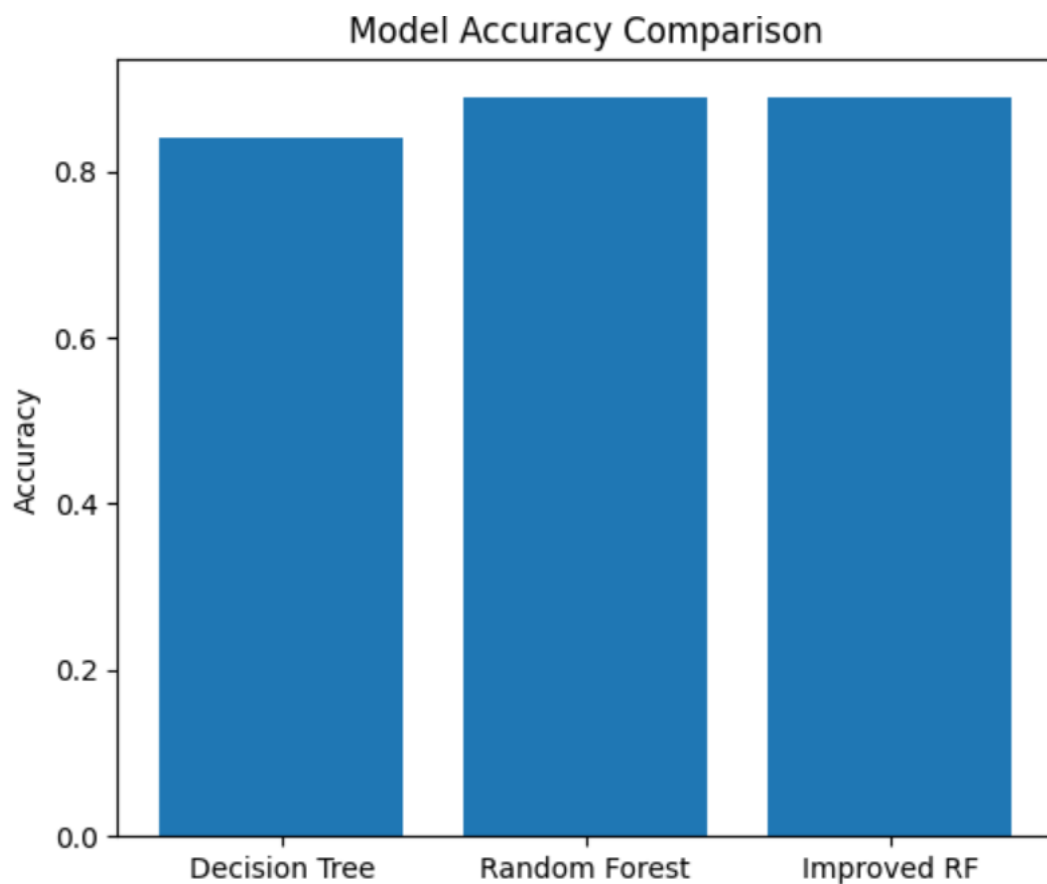
3. Improved Random Forest

The Random Forest model was further optimized by adjusting hyperparameters such as:

- Number of trees (n_estimators)
- Tree depth (max_depth)
- Balanced class weighting

Accuracy Achieved: 89%

The improved model achieved the same accuracy as the standard Random Forest model, indicating that the original Random Forest model was already highly optimized.



Results & Discussion

The performance of all machine learning models is shown below:

Model	Accuracy
Decision Tree:	84%
Random Forest::	89%

Model	Accuracy
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Improved Random Forest:	89%
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The results indicate that Random Forest and Improved Random Forest outperformed the Decision Tree model.

The Decision Tree model provided good performance but was less stable compared to Random Forest. Random Forest produced better predictions because it combines multiple trees, reducing overfitting and improving generalization.

The Improved Random Forest model did not significantly improve performance because the baseline Random Forest model was already producing strong results.

Confusion Matrix Explanation

A confusion matrix was used to evaluate the performance of the classification model.

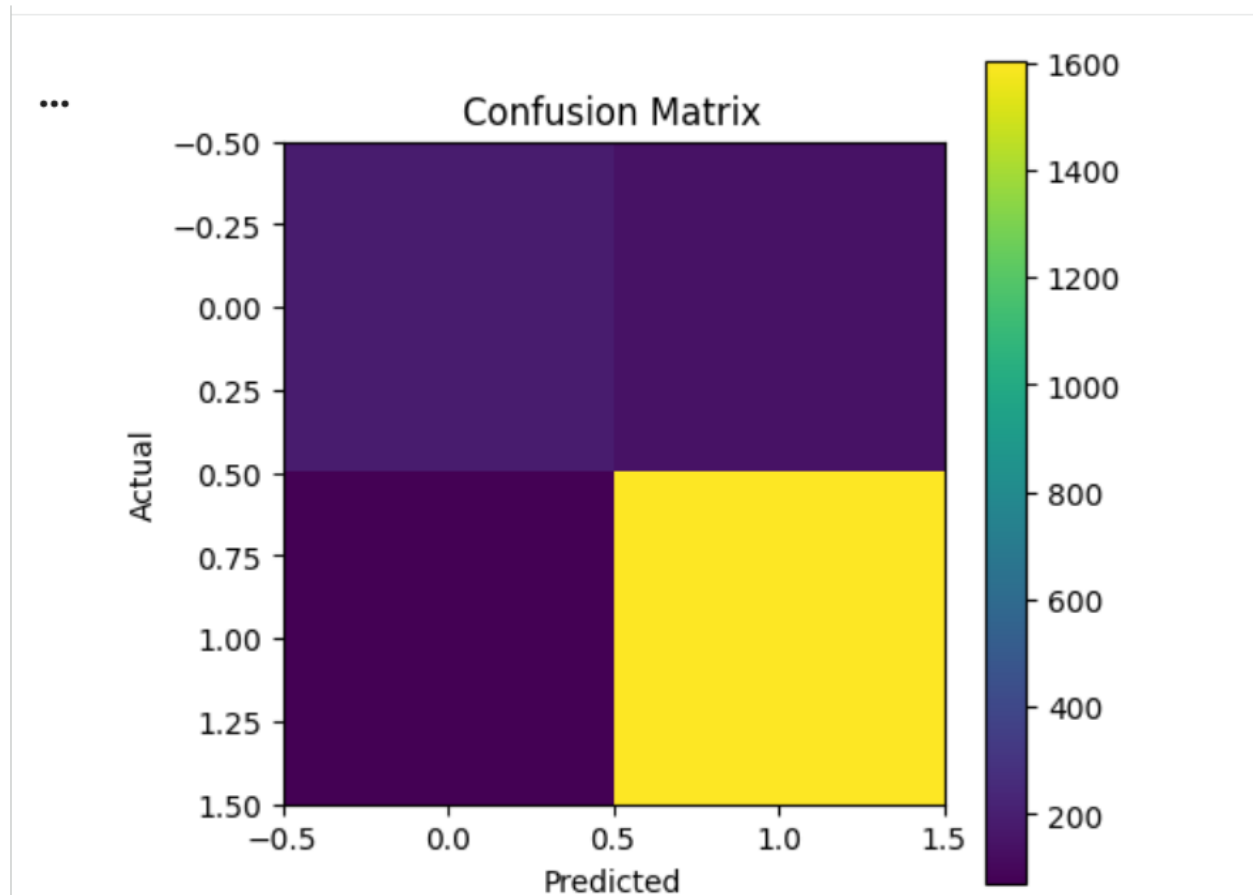
The confusion matrix helps identify:

- Correct predictions
- Incorrect predictions
- True positives
- True negatives
- False positives
- False negatives

In this project, the confusion matrix showed that the model correctly classified the majority of placed and not placed students, indicating strong predictive performance.

The high number of correct predictions demonstrates that the model successfully learned patterns from student academic and behavioral data.

Here is graphical representation:



Classification Report Explanation

The classification report was used to evaluate the model in detail using the following metrics:

Precision

Precision measures how many predicted positive cases were actually correct.

Recall

Recall measures how many actual positive cases were correctly identified.

F1-Score

F1-score provides a balance between precision and recall.

Accuracy

Accuracy measures the overall correctness of the model.

The classification report indicated strong model performance, with Random Forest achieving high predictive accuracy and balanced performance across classes.

Conclusion

This project successfully applied machine learning techniques to predict student placement status using academic and behavioral factors. Three machine learning models were implemented and compared.

The Random Forest and Improved Random Forest models achieved the highest accuracy of **89%**, outperforming the Decision Tree model, which achieved **84% accuracy**.

The results demonstrate that machine learning can effectively predict placement outcomes and assist educational institutions in identifying students who may require additional academic support to improve employability.

Future Scope

This project can be improved in the future through:

1. Using larger and more diverse datasets.
2. Applying advanced machine learning algorithms such as XGBoost and Neural Networks.
3. Including additional student-related features such as communication skills and extracurricular activities.
4. Deploying the model as a web application for real-time placement prediction.
5. Integrating predictive analytics into educational systems to support student career planning.

Link: <https://colab.research.google.com/drive/1LqBvnIG8uXy5RZJ4LfdgmFM639eGkxVJ>

Task#2

Building a Simple AI Agent Web App

Abstract

This project presents the development of a **Student AI Assistant** using the **Groq API** and **Streamlit** framework. The purpose of this AI-powered assistant is to help students with study-related questions by providing simple explanations of academic concepts. The assistant uses a large language model (LLM) to understand user questions and generate intelligent responses. The application provides an easy-to-use interface where students can ask questions related to

programming, artificial intelligence, mathematics, and general educational topics. The project demonstrates the integration of an API, prompt engineering, and web application development.

Introduction

Artificial Intelligence (AI) has transformed education by making learning more interactive and accessible. AI-powered assistants can support students by answering questions, explaining difficult topics, and improving understanding. In this project, a **Student AI Assistant** was developed to provide educational support using the **Groq API**.

The assistant is designed to answer only study-related questions and guide students in learning concepts in a simple and friendly manner. The project was implemented as a web application using **Streamlit**, which provides a user-friendly interface for interaction.

Problem Statement

Students often face difficulties in understanding complex academic topics and may not always have access to immediate guidance. Traditional learning methods can sometimes be time-consuming and lack personalized assistance. Therefore, there is a need for an intelligent educational assistant that can provide instant academic help.

The main problem addressed in this project is:

“How can an AI-powered assistant help students understand educational concepts and answer study-related questions effectively?”

Objectives

The objectives of this project are:

- To build an AI-powered student assistant.
- To integrate the **Groq API** with a Python application.
- To develop a simple and interactive web interface.
- To provide educational support for students.
- To answer study-related questions in simple language.
- To improve learning through instant AI-generated explanations.

Technologies Used

The following technologies and tools were used in this project:

Python: Backend programming

Streamlit: Web application framework

Groq API: AI model integration

Python-dotenv: Secure API key management

Llama Model/: Generating AI responses

Methodology

The development of the Student AI Assistant followed these steps:

1. Environment Setup

Necessary Python libraries such as **Streamlit**, **Groq**, and **python-dotenv** were installed using pip.

2. API Integration

A free API key was generated from the Groq platform and connected securely through a .env file to avoid hardcoding sensitive information.

3. System Prompt Design

A system prompt was implemented to define the role of the assistant. The assistant was instructed to behave as a helpful educational tutor and respond only to study-related queries.

4. User Interface Development

A web interface was developed using Streamlit, allowing users to type questions and receive AI-generated responses instantly.

5. Response Generation

When a student enters a question, the input is sent to the Groq API, processed by the language model, and an intelligent response is returned to the user.

API Integration

The Groq API was integrated into the project to provide intelligent responses using a language model. The API key was stored securely in the .env file.

The model used in this project was:

llama-3.1-8b-instant

This model was selected because it provides fast and efficient responses suitable for educational purposes.

System Prompt

The Student AI Assistant was designed with a specific system role to ensure educational behavior.

The assistant was instructed to:

- Help students understand concepts.
- Explain topics in simple language.
- Answer educational questions.
- Politely avoid unrelated discussions.
- Maintain a friendly and helpful tone.

This system prompt improved the consistency and accuracy of responses.

Web Interface

The application contains a simple and user-friendly interface. The interface includes:

- A title for the application (**Student AI Assistant**)
- A text input box for entering questions
- A submit button
- A response section displaying AI-generated answers

The interface was created using Streamlit to ensure simplicity and ease of use. Interface is shown in image below

Student AI Assistant

Ask me study-related questions!

Enter your question:

Ask AI



Testing and Results

The Student AI Assistant was tested using multiple educational questions to evaluate its performance.

Example Test Questions:

- What is Machine Learning?
- Explain Python loops.
- What is Artificial Intelligence?
- Explain confusion matrix.
- What is supervised learning?

Results:

The AI assistant successfully generated relevant and understandable responses for study-related questions. It provided quick explanations and maintained an educational focus throughout the interaction. As show in image:



Student AI Assistant

Ask me study-related questions!

Enter your question:

what is ai

Ask AI

AI Response

What is AI?

Artificial Intelligence (AI) is a type of computer science that enables machines to think and behave like humans. It's like a superpower that helps computers learn, reason, and solve problems on their own!

Imagine you're playing a game of chess with a friend. You think about the next move, analyze the board, and try to outsmart your opponent. AI works in a similar way. It uses algorithms, data, and rules to make decisions, recognize patterns, and adapt to new situations.

Types of AI:

Challenges Faced

During development, some issues were encountered:

- Package installation errors.
- API integration difficulties.
- Model deprecation issue with the old Groq model.
- Configuration of environment variables.

These problems were solved by updating dependencies and using the latest supported Groq model.

Conclusion

In conclusion, the **Student AI Assistant** was successfully developed using the **Groq API** and **Streamlit**. The project achieved its objective of creating an educational AI assistant capable of answering academic questions. The assistant provides a simple and effective learning experience for students and demonstrates the practical use of AI in education.

Future Scope

This project can be improved in the future by adding:

- Voice interaction support.
- Multiple subject specialization.
- Chat history functionality.
- Better user interface design.
- Support for file and PDF-based question answering.
- Multi-language support for students.

Python code:

```
import streamlit as st
from groq import Groq
from dotenv import load_dotenv
import os

# Load environment variables
load_dotenv()

# Get API key
api_key = os.getenv("GROQ_API_KEY")

# Initialize Groq client
client = Groq(api_key=api_key)

# Streamlit UI
st.set_page_config(page_title="Student AI Assistant")

st.title("🎓 Student AI Assistant")
st.write("Ask me study-related questions!")

# User input
user_question = st.text_input("Enter your question:")

# Button
if st.button("Ask AI"):

    if user_question:

        # System prompt
        system_prompt = """
        You are a helpful Student AI Assistant.
        Your job is to help students understand concepts,
        explain topics simply, solve study-related questions,
        and guide them in learning.

        Rules:
        - Explain in simple language.
        - Help with assignments conceptually.
        - Be friendly and educational.
        - If asked unrelated things, politely redirect to studies.
        """

        response = client.chat.completions.create(
            model="llama-3.1-8b-instant",
            messages=[
                {"role": "system", "content": system_prompt},
                {"role": "user", "content": user_question}
            ]
        )
```

```
    answer = response.choices[0].message.content

    st.subheader("AI Response")
    st.write(answer)

else:
    st.warning("Please enter a question.")
```